

Basic Networking Commands

Ex No: 1

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Aim: study of various network commands used in linux and windows

1) arp -a:

Display the address Resolution Protocol (ARP) table, which contains the IP addresses and their corresponding MAC (physical) address stored in the system.

Output:

Interface: 192.168.56.1 --- 0xc

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	Static
224.0.0.2	01-00-5e-00-00-02	Static
224.0.0.22	01-00-5e-00-00-1b	Static

2) Hostname:

Display the hostname (computer name) of the local machine.

Output:

DESKTOP-VTBNKNE

3) ipconfig /all

Displays detailed information about the network adapter including IP address, subnet mask, default gateway, DNS Server, MAC address, etc.

Output :-

Windows IP Configuration
Host Name DESKTOP-K7BNKNE
Primary Dns Suffix
Node Type Mixed
IP Routing Enabled No

4) netstat -a

Displays the NetPLOS name table of a remote or local computer and help troubles shoot NetBios name resolution problems.

Output :

Ethernet :
Node IP Address : [192.168.56.1] Scope Id : [3]
Host not found
Hotspotshield
Node IP Address : [0.0.0.0] Scope Id : [3]
Host not found

5) netstat -r : netstat -r : netstat -r (F)

Displays active TCP/UDP connections,
listening ports, routing table and network
statistics

output :

Interface List

9...00 ff ac ba 12 06... Hotspotshield TAP-Windos
Adapter V9
10...0a 00 27 00 00 0a... VirtualBox Host-Only
Ethernet Adapter
16...02 e9 aa a1 59 87... Microsoft Wi-Fi
Direct Virtual
Adapter
19...06 e9 aa a1 59 87... Microsoft Wi-Fi
Direct Virtual
Adapter #2

6) nslookup google.com

Queries the DNS server to obtain
the IP address of the domain name.

Output :

Server : LAPTOP-38UHQ TAB

Address : 192.168.1.1

Non-authoritative answer:

Name : google.com

Address : 216.239.39.111

142.250.67.46

7) Pathping google.com :

Combines the features of Tracer and Traceroute to trace the path to a destination and Calculate packet loss at each hop.

Output :

Tracing route to google.com [142.250.67.46] over a maximum of 30 hops :

0 DESKTOP-A7BNKNE.mshome.net [192.168.137.150]

1 LAPTOP-J8UHT4B.mshome.net [192.168.137.17]

2 ...

8) Ping google.com :

checks whether/another computer or website is reachable over the networking using ICMP Request and Echo reply messages.

Output :

Pinging google.com [142.250.67.46] with 32 bytes of data :

Reply from 142.250.67.46 : bytes=32 time=10 ms TTL=117

Reply from 142.250.67.46 : bytes=32 time=7 ms TTL=117

Reply from 142.250.67.46: bytes=32 time=14 ms

TTL

Reply from 142.250.67.46: bytes=32 time=18 ms

TTL = 177

9) route

Displays or modifies the IP routing

table used by the operating system.

Output:

manipulates network routing tables.

ROUTE [-F] [-P] [-A] [-6] command [destination]
[network] [gateway] [METRIC metric]
[IF interface]

-F

clears the routing tables of all

gateway entries. If this is used in

conjunction with one of the commands,

the tables are cleared prior to

running the command.

LINUX COMMANDS

10) ip address show

Displays and messages IP address,
routing and network interfaces.

Output:-

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536  
qdisc noop state UNKNOWN group default  
link/loopback 00:00:00:00:00:00 brd
```

inet 172.17.0.1/8 scope host 16

Valid: 256 Jitter: preferred: 256 Jitter: 256

11) ifconfig

Display network interface details such as IP address and network statistics

Output:

eth0: flags=4163<UP,BROADCAST,PROMISC

MULTICAST>mtu 1500

inet 172.17.197.94 netmask 255.255

240.0 broadcast 172.17.255.255

inet 6 fe80::215:5dff:fe82:de86

Prefixlen 64 Scopeid 0x20<link>

13) mttr google.com :-

Combines Ping and Traceroute to display the network path and performance statistics

Output:

DESKTOP

LAPTOP-130H0TAB (172.17.197.94) -> google.com

Host

	Loss%	Sent	Last
1. DESKTOP-K7BNKNE mihomc.net	0.0%	128	0.7
2. 192.168.1.1	0.0%	141	4.7
3. 100.84.0.1	0.0%	163	5.7
4. (waiting for reply)	0.0%		

- 13) tcpdump -i eth0
Captures and displays network
packets passing through a network interface
for analysis

output:

tcpdump: verbose output suppressed, use -v or
-vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet)
capture size 2048 bytes
10:15:21.123456 IP 192.168.1.10.52124 → 8.8.8.8.53: UDP, length 32

14)

Ping

google.com

Sends ICMP Echo request packets to
verify network connectivity and measure
response time.

output:

PING google.com (142.251.43.206) 56(84) bytes
64 bytes from 1cmce0a-bb-in-f14.1e400.net
(142.251.43.206): icmp_seq=1 ttl=117
time=5.76 ms

Result:

Thus the various windows and linux
networking commands were executed
successfully.